

# Supplementary material — Unknown is not normal

## S1. Calculator readings awaiting clinical review

Each calculator cites its primary source in code. Where the source was ambiguous or differed from common implementations, the implemented reading and the alternatives are listed below. The author, an emergency physician, reviewed and accepted the implemented readings.

### Type:

- **S (source):** resolvable by reading the full text of the primary paper. These cluster on the abstract-only [A] sources, PERC and Wells 2000.
- **C (clinical):** needs clinical judgement.
- **M (modelling):** a modelling choice with no clinical answer.

**Priority** is face validity: how much a clinician reading a generated case would notice if it were wrong.

- **High:** visible in every case of that calculator.
- **Med:** visible in some cases.
- **Low:** internal only.

For a bounds benchmark most items only need to be *fixed and documented*: the ground truth is self-consistent either way.

| # | Type | Priority | Calculator | Item           | Implemented                                                                                                                                      | Alternatives / notes                                                                                                                      |
|---|------|----------|------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | S    | High     | HEART      | Troponin bands | $\leq$ normal / $1-3\times / > 3\times$ , deliberately following the validation cohorts (Backus 2010/2013, Poldervaart 2013), not the 2008 table | Six 2008: $1-2\times / > 2\times$ . Exactly $3\times$ at the band edge: $\geq 3\times$ (Backus 2011, Poldervaart) vs $> 3\times$ (MDCalc) |

| #  | Type | Priority | Calculator | Item                     | Implemented                                                                                                                                                                                                             | Alternatives / notes                                                                                              |
|----|------|----------|------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 27 | C    | High     | HEART      | Troponin assay reference | Troponin is a level relative to the assay's upper reference limit, accepted by the author. Checked: every HEART note states the level relative to the reference limit; the one raw value also gives the reference limit | Real hs-troponin uses sex-specific 99th percentiles. Out of scope while troponin is a ratio; needed for real data |
| 23 | S    | High     | HEART      | Age exactly 65           | 2 points (age $\geq 65$ ), per the 2008 text ("65 years or older")                                                                                                                                                      | 2008 table says " $\leq 65$ " (typo); some renderings use $45-65 \square 1$ , $> 65 \square 2$ (65 scores 1)      |
| 2  | S    | High     | HEART      | Age exactly 45           | 1 point (age $\geq 45$ )                                                                                                                                                                                                | Poldervaart 2013: $\leq 45 \square 0$                                                                             |
| 4  | C    | High     | HEART      | Smoking recency          | $\leq 3$ months (MDCalc), accepted by the author                                                                                                                                                                        | $< 1$ month (Six 2008). No note states a quit date, so the choice affects no case                                 |
| 5  | C    | High     | HEART      | Family history           | first-degree relative with CAD before 65, accepted by the author                                                                                                                                                        | Only 3 notes give an age (mother 54, brother 55, father 58), all clear-cut, so the choice affects no case         |
| 6  | S    | Med      | HEART      | Obesity                  | boolean, labelled BMI $> 30$                                                                                                                                                                                            | not defined in Six 2008; BMI $> 30$ in Poldervaart 2013                                                           |

| #  | Type | Priority | Calculator | Item                           | Implemented                                                        | Alternatives / notes                                                                                                                       |
|----|------|----------|------------|--------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | S    | Med      | HEART      | TIA as atherosclerotic disease | included (2 points)                                                | Six 2008 lists revascularisation, MI, stroke, PAD. MedCalc-Bench counts TIA as an ordinary risk factor (1 of 100 anchor disagreements)     |
| 3  | S    | Med      | HEART      | ECG 2 points                   | significant ST deviation (depression or elevation)                 | 2008 table: ST depression only                                                                                                             |
| 11 | S    | High     | PERC       | SaO2                           | positive if < 95%, on room air (the renderer states "on room air") | 2004 abstract: rule needs > 94%. "Sats normal on 2 L" is normal but PERC-positive in some readings; our notes never render supplemental O2 |
| 12 | S    | Med      | PERC       | Surgery/trauma                 | within 4 weeks requiring hospitalisation (2008 abstract)           | requiring general anaesthesia (some sources)                                                                                               |
| 13 | S    | Med      | PERC       | Hormone use                    | exogenous oestrogen                                                | any hormone therapy (2004 abstract: "hormone use")                                                                                         |

| #  | Type | Priority | Calculator | Item                    | Implemented                                                                                                                                                                                                   | Alternatives / notes                                                                 |
|----|------|----------|------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 24 | S    | Med      | Wells      | Item definitions        | Stated in the parameter labels given to renderer and extractor: malignancy = on treatment, treated within 6 months, or palliative; DVT signs = leg swelling plus pain on deep-vein palpation; previous DVT/PE | Previous DVT/PE should say “objectively diagnosed” (label says “Previous DVT or PE”) |
| 14 | S    | Med      | Wells      | Immobilisation duration | “> 3 days” (NICE)                                                                                                                                                                                             | “≥ 3 days” (MDCalc)                                                                  |
| 25 | S    | Med      | Wells      | 4-week window           | label: “Immob-<br>ilisation > 3 days or surgery in the previous 4 weeks” (window read as applying to both)                                                                                                    | the window may apply to surgery only                                                 |
| 15 | S    | Med      | Wells      | PE-likely item          | “alternative diagnosis less likely than PE” (NICE)                                                                                                                                                            | “PE #1 or equally likely” (MDCalc)                                                   |
| 8  | C    | Med      | CURB-65    | BUN equivalent          | code converts BUN to urea mmol/L and applies > 7 (= BUN > 19.6)                                                                                                                                               | MDCalc: BUN > 19; an audit suggests > 20                                             |

| #  | Type | Priority | Calculator      | Item                                                        | Implemented                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Alternatives / notes                                          |
|----|------|----------|-----------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 26 | C    | High     | CURB-65 / qSOFA | How mental status is extracted (accepted: GCS 14 □ present) | <p>Bool params, labelled “Confusion (AMT <math>\leq</math> 8 or new disorientation)” and “Altered mentation (GCS &lt; 15)”. There is no GCS parameter: “GCS 14” □ altered mentation present; a vague “drowsy” is left to the extractor’s judgement. Checked: dementia, baseline impairment and vague words (“lethargic”) appear only where mental status is explicitly documented, never where it is undocumented, so no case is ambiguous. Baseline impairment is consistent with each definition (qSOFA counts any abnormal mentation; CURB-65 counts only new disorientation)</p> | add a numeric GCS parameter (would change prompts and cohort) |

| #  | Type | Priority | Calculator      | Item                           | Implemented                         | Alternatives / notes                                                                                                               |
|----|------|----------|-----------------|--------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 9  | C    | Low      | CURB-65 / qSOFA | Confusion vs altered mentation | two separate parameters             | Coupling is not needed: each case belongs to exactly one calculator, so one patient never carries both                             |
| 10 | C    | Med      | qSOFA           | Altered mentation threshold    | GCS < 15 (Sepsis-3)                 | GCS ≤ 13 (Seymour derivation model)                                                                                                |
| 16 | C    | Med      | Cockcroft-Gault | Weight                         | actual body weight (original)       | ideal/adjusted weight in obesity; MedCalc-Bench picks by BMI (explains 49% anchor disagreement)                                    |
| 17 | M    | Low      | Cockcroft-Gault | Decision thresholds            | 30, 60 mL/min (config-overrideable) | FDA 2024 Table 1 also has 90                                                                                                       |
| 18 | M    | Low      | Units           | Creatinine factor              | 88.4 μmol/L per mg/dL               | 88.42 (10,000 / 113.12). Clinically irrelevant; can move a boundary case by < 0.03% of CrCl. No case sits that close in our cohort |
| 19 | M    | Med      | Params          | “Stated normal” intervals      | see table below                     | —                                                                                                                                  |

| #  | Type | Priority | Calculator | Item                         | Implemented             | Alternatives / notes                                                                                                                                                                                                                                                                                                           |
|----|------|----------|------------|------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | M    | Low      | Params     | Creatinine can't be "normal" | Present or Unknown only | The no-straddle guarantee is per parameter and threshold. CG thresholds apply to a function of age, weight and creatinine, so no creatinine interval could guarantee a single CrCl band. For CG, bounds return a set of categories whenever an input is uncertain (100 of 200 CG cases are determined from the note by design) |

| #  | Type | Priority | Calculator | Item              | Implemented                                | Alternatives / notes                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------|----------|------------|-------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 | M    | Med      | Cohort     | Population priors | independent sampling (plus DBP < SBP – 15) | Independence can yield implausible combinations (e.g. established CAD with no risk factors, or age < 40 with ≥ 3 risk factors: 12 HEART cases, 6% of HEART and 1% of the cohort). It doesn't affect scores (atherosclerotic disease scores 2 regardless), and excluding the flagged cases leaves results unchanged (supplementary Table S10). Reported as a limitation; correlated priors are future work |



| #  | Type | Priority | Calculator | Item                              | Implemented                                                                                                                                                                                                                                                                                                                                                                                                               | Alternatives / notes                                    |
|----|------|----------|------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 22 | C    | Med      | Cohort     | Medication implying a comorbidity | Only metformin <input type="checkbox"/> diabetes is reasonably specific.<br>Lisinopril <input type="checkbox"/> hypertension and atorvastatin <input type="checkbox"/> hypercholesterolaemia are weak (ACE inhibitors in HF, post-MI and CKD; statins post-ACS and in diabetes).<br><b>Scoring:</b> for a comorbidity shown only through a medication, both Present and Unknown count as correct; only Absent is an error | treat the implication as probabilistic in the generator |

## S2. Population priors for the synthetic cohort

Plausible emergency-department populations for each calculator's intended use; parameters sampled independently except diastolic < systolic blood pressure – 15 mmHg. The PERC cohort represents low gestalt pre-test probability. Reviewed by the author.

| Calculator | Parameter            | Prior                                                                             |
|------------|----------------------|-----------------------------------------------------------------------------------|
| HEART      | heart_history        | slightly_suspicious 0.45,<br>moderately_suspicious 0.35,<br>highly_suspicious 0.2 |
| HEART      | heart_ecg            | normal 0.6,<br>nonspecific_repolarization 0.3,<br>significant_st_deviation 0.1    |
| HEART      | age                  | Normal(58, 15) truncated to [18, 100]                                             |
| HEART      | hypertension         | P(yes) = 0.4                                                                      |
| HEART      | hypercholesterolemia | P(yes) = 0.3                                                                      |

| Calculator      | Parameter                 | Prior                                                 |
|-----------------|---------------------------|-------------------------------------------------------|
| HEART           | diabetes                  | P(yes) = 0.2                                          |
| HEART           | obesity                   | P(yes) = 0.25                                         |
| HEART           | smoking                   | P(yes) = 0.25                                         |
| HEART           | family_history_cad        | P(yes) = 0.2                                          |
| HEART           | atherosclerotic_disease   | P(yes) = 0.2                                          |
| HEART           | heart_troponin            | le_normal 0.75, 1_to_3x_normal 0.15, gt_3x_normal 0.1 |
| CURB-65         | confusion                 | P(yes) = 0.15                                         |
| CURB-65         | urea                      | Normal(7, 3.5) truncated to [1.5, 40]                 |
| CURB-65         | resp_rate                 | Normal(22, 6) truncated to [8, 50]                    |
| CURB-65         | sbp                       | Normal(125, 25) truncated to [60, 220]                |
| CURB-65         | dbp                       | Normal(72, 14) truncated to [30, 130]                 |
| CURB-65         | age                       | Normal(65, 17) truncated to [18, 100]                 |
| qSOFA           | resp_rate                 | Normal(21, 5) truncated to [8, 50]                    |
| qSOFA           | altered_mentation         | P(yes) = 0.2                                          |
| qSOFA           | sbp                       | Normal(118, 24) truncated to [60, 220]                |
| PERC            | age                       | Normal(38, 12) truncated to [18, 90]                  |
| PERC            | heart_rate                | Normal(88, 15) truncated to [45, 160]                 |
| PERC            | spo2                      | Normal(97, 2) truncated to [85, 100]                  |
| PERC            | unilateral_leg_swelling   | P(yes) = 0.05                                         |
| PERC            | hemoptysis                | P(yes) = 0.03                                         |
| PERC            | recent_surgery_trauma     | P(yes) = 0.05                                         |
| PERC            | prior_vte                 | P(yes) = 0.05                                         |
| PERC            | hormone_use               | P(yes) = 0.12                                         |
| Wells PE        | dvt_signs                 | P(yes) = 0.12                                         |
| Wells PE        | pe_most_likely            | P(yes) = 0.3                                          |
| Wells PE        | heart_rate                | Normal(95, 18) truncated to [45, 170]                 |
| Wells PE        | immobilization_or_surgery | P(yes) = 0.12                                         |
| Wells PE        | prior_vte                 | P(yes) = 0.12                                         |
| Wells PE        | hemoptysis                | P(yes) = 0.05                                         |
| Wells PE        | malignancy                | P(yes) = 0.1                                          |
| Cockcroft-Gault | age                       | Normal(60, 17) truncated to [18, 100]                 |
| Cockcroft-Gault | weight                    | Normal(75, 17) truncated to [35, 180]                 |
| Cockcroft-Gault | creatinine                | Normal(1.3, 0.8) truncated to [0.4, 8]                |
| Cockcroft-Gault | sex                       | male 0.5, female 0.5                                  |

### S3. Missingness sensitivity (oracle extraction, natural share of undetermined cases)

Fresh cohorts (200 cases per calculator) at each missingness level; stratum results reweighted to the natural share of undetermined cases.

| Missingness | System                     | Accuracy (natural share) | Questions/case (natural share) | Undetermined share |
|-------------|----------------------------|--------------------------|--------------------------------|--------------------|
| 0.1         | S1 Ask-all                 | 0.999                    | 0.642                          | 0.252              |
| 0.1         | S3-bin<br>(missing=normal) | 0.964                    | 0.089                          | 0.252              |
| 0.1         | S3 Bounds                  | 0.999                    | 0.326                          | 0.252              |

| Missingness | System           | Accuracy (natural share) | Questions/case (natural share) | Undetermined share |
|-------------|------------------|--------------------------|--------------------------------|--------------------|
| 0.1         | S4               | 0.999                    | 0.316                          | 0.252              |
|             | Bounds+VOI+echo  |                          |                                |                    |
| 0.3         | S1 Ask-all       | 0.997                    | 1.925                          | 0.61               |
| 0.3         | S3-bin           | 0.899                    | 0.282                          | 0.61               |
|             | (missing=normal) |                          |                                |                    |
| 0.3         | S3 Bounds        | 0.997                    | 1.144                          | 0.61               |
| 0.3         | S4               | 0.997                    | 1.082                          | 0.61               |
|             | Bounds+VOI+echo  |                          |                                |                    |
| 0.5         | S1 Ask-all       | 0.994                    | 3.289                          | 0.824              |
| 0.5         | S3-bin           | 0.824                    | 0.483                          | 0.824              |
|             | (missing=normal) |                          |                                |                    |
| 0.5         | S3 Bounds        | 0.994                    | 2.159                          | 0.824              |
| 0.5         | S4               | 0.994                    | 1.979                          | 0.824              |
|             | Bounds+VOI+echo  |                          |                                |                    |

#### S4. Confidence-echo threshold sweep (S4 policy, clean notes, ideal clinician)

| extraction         | echo_threshold | accuracy | mean_questions | echo_per_case | echo_caught_errors |
|--------------------|----------------|----------|----------------|---------------|--------------------|
| haiku__sonnet      | 0.5            | 0.995    | 0.8833         | 0.005         | 5                  |
| haiku__sonnet      | 0.8            | 0.9967   | 0.895          | 0.015         | 9                  |
| haiku__sonnet      | 0.9            | 0.9967   | 0.895          | 0.015         | 9                  |
| haiku__sonnet      | 0.95           | 0.9967   | 0.895          | 0.015         | 9                  |
| haiku__sonnet      | 0.99           | 0.9983   | 1.705          | 0.8242        | 19                 |
| haiku__sonnet      | 0.999          | 0.9983   | 1.7075         | 0.8267        | 19                 |
| qwen_local__sonnet | 0.5            | 0.9983   | 1.1783         | 0.0           | 0                  |
| qwen_local__sonnet | 0.8            | 0.9983   | 1.1783         | 0.0           | 0                  |
| qwen_local__sonnet | 0.9            | 0.9983   | 1.1983         | 0.02          | 2                  |
| qwen_local__sonnet | 0.95           | 0.9983   | 1.2967         | 0.1183        | 5                  |
| qwen_local__sonnet | 0.99           | 0.9983   | 1.8142         | 0.6358        | 6                  |
| qwen_local__sonnet | 0.999          | 0.9983   | 1.8142         | 0.6358        | 6                  |

#### S5. Calibration of extraction confidence

Present/absent claims; isotonic and temperature calibration fitted on a seeded 30% development split, evaluated on the rest. Haiku 4.5 (self-reported): ECE 0.014 raw, 0.003 temperature, 0.002 isotonic. Qwen3.5-9B (token log-probabilities): ECE 0.048 raw, 0.033 temperature, 0.002 isotonic.

#### S6. Full results grid (559 paired cases; both extractors, both note styles, both clinicians)

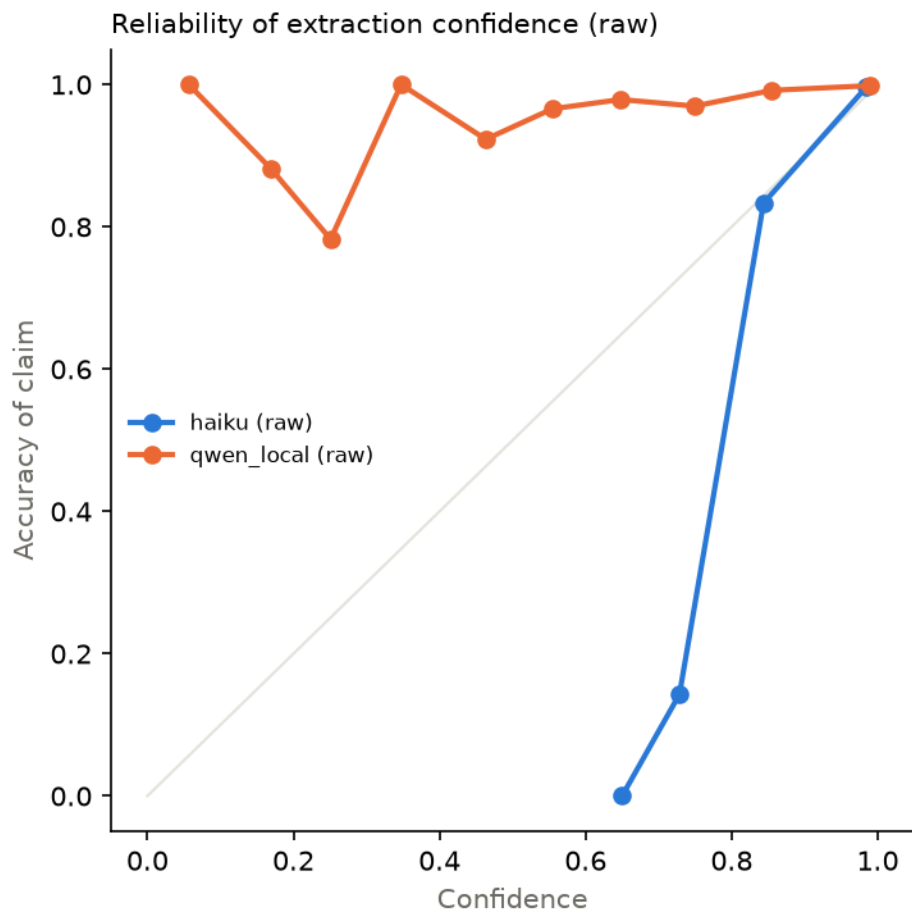


Figure 1: Reliability of raw extraction confidence (test split).

| Condition                                       | System                          | n   | Accuracy,<br>% (95%<br>CI) | Correct<br>when an-<br>swering,<br>% | Under-<br>triage, %<br>(95% CI) | Over-<br>triage, % | Premature<br>commit-<br>ment, % | Questions/cases-<br>(95% CI) | Irrelevant<br>cases-<br>tions, % |
|-------------------------------------------------|---------------------------------|-----|----------------------------|--------------------------------------|---------------------------------|--------------------|---------------------------------|------------------------------|----------------------------------|
| Haiku,<br>clean<br>notes,<br>ideal<br>clinician | S1<br>Ask-all                   | 559 | 99.1<br>(97.9–<br>99.6)    | 99.3                                 | 0.2<br>(0.0–1.0)                | 0.7                | 0.0                             | 1.71<br>(1.60–<br>1.81)      | 50.0                             |
| Haiku,<br>clean<br>notes,<br>ideal<br>clinician | S2 LLM<br>agent                 | 559 | 99.6<br>(98.7–<br>99.9)    | 99.8                                 | 0.2<br>(0.0–1.0)                | 0.2                | 0.2                             | 0.92<br>(0.82–<br>1.01)      | 10.1                             |
| Haiku,<br>clean<br>notes,<br>ideal<br>clinician | S3<br>Bounds                    | 559 | 99.1<br>(97.9–<br>99.6)    | 99.3                                 | 0.2<br>(0.0–1.0)                | 0.7                | 0.0                             | 0.85<br>(0.75–<br>0.94)      | 0.0                              |
| Haiku,<br>clean<br>notes,<br>ideal<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 99.6<br>(98.7–<br>99.9)    | 99.8                                 | 0.2<br>(0.0–1.0)                | 0.2                | 0.0                             | 0.82<br>(0.73–<br>0.91)      | 0.0                              |
| Haiku,<br>clean<br>notes,<br>ideal<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 91.4<br>(88.8–<br>93.5)    | 91.4                                 | 8.2 (6.2–<br>10.8)              | 0.4                | 37.0                            | 0.20<br>(0.16–<br>0.24)      | 0.0                              |
| Haiku,<br>clean<br>notes,<br>noisy<br>clinician | S1<br>Ask-all                   | 559 | 88.2<br>(85.3–<br>90.6)    | 97.6                                 | 0.5<br>(0.2–1.6)                | 8.8                | 0.0                             | 1.71<br>(1.60–<br>1.81)      | 48.6                             |
| Haiku,<br>clean<br>notes,<br>noisy<br>clinician | S2 LLM<br>agent                 | 559 | 84.4<br>(81.2–<br>87.2)    | 97.7                                 | 0.5<br>(0.2–1.6)                | 13.1               | 1.8                             | 0.95<br>(0.84–<br>1.05)      | 8.5                              |
| Haiku,<br>clean<br>notes,<br>noisy<br>clinician | S3<br>Bounds                    | 559 | 88.2<br>(85.3–<br>90.6)    | 97.6                                 | 0.5<br>(0.2–1.6)                | 8.8                | 0.0                             | 0.88<br>(0.77–<br>0.97)      | 0.0                              |
| Haiku,<br>clean<br>notes,<br>noisy<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 88.7<br>(85.8–<br>91.1)    | 98.0                                 | 0.5<br>(0.2–1.6)                | 8.2                | 0.0                             | 0.86<br>(0.76–<br>0.96)      | 0.0                              |

| Condition                                       | System                          | n   | Accuracy,<br>% (95%<br>CI) | Correct<br>when an-<br>swering,<br>% | Under-<br>triage, %<br>(95% CI) | Over-<br>triage, % | Premature<br>commit-<br>ment, % | Questions/cases-<br>(95% CI) | Irrelevant<br>cases-<br>tions, % |
|-------------------------------------------------|---------------------------------|-----|----------------------------|--------------------------------------|---------------------------------|--------------------|---------------------------------|------------------------------|----------------------------------|
| Haiku,<br>clean<br>notes,<br>noisy<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 88.7<br>(85.8–<br>91.1)    | 91.2                                 | 8.2 (6.2–<br>10.8)              | 2.5                | 36.0                            | 0.20<br>(0.16–<br>0.25)      | 0.0                              |
| Haiku,<br>messy<br>notes,<br>ideal<br>clinician | S1<br>Ask-all                   | 559 | 98.6<br>(97.2–<br>99.3)    | 98.7                                 | 0.5<br>(0.2–1.6)                | 0.9                | 0.0                             | 1.92<br>(1.80–<br>2.04)      | 45.6                             |
| Haiku,<br>messy<br>notes,<br>ideal<br>clinician | S2 LLM<br>agent                 | 559 | 99.8<br>(99.0–<br>100.0)   | 100.0                                | 0.0<br>(0.0–0.7)                | 0.2                | 0.7                             | 0.92<br>(0.82–<br>1.01)      | 10.7                             |
| Haiku,<br>messy<br>notes,<br>ideal<br>clinician | S3<br>Bounds                    | 559 | 98.6<br>(97.2–<br>99.3)    | 98.7                                 | 0.5<br>(0.2–1.6)                | 0.9                | 0.0                             | 1.04<br>(0.94–<br>1.15)      | 0.0                              |
| Haiku,<br>messy<br>notes,<br>ideal<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 99.1<br>(97.9–<br>99.6)    | 99.3                                 | 0.5<br>(0.2–1.6)                | 0.4                | 0.0                             | 1.04<br>(0.94–<br>1.15)      | 0.0                              |
| Haiku,<br>messy<br>notes,<br>ideal<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 90.7<br>(88.0–<br>92.8)    | 90.7                                 | 8.8 (6.7–<br>11.4)              | 0.5                | 39.4                            | 0.30<br>(0.23–<br>0.36)      | 0.0                              |
| Haiku,<br>messy<br>notes,<br>noisy<br>clinician | S1<br>Ask-all                   | 559 | 85.5<br>(82.3–<br>88.2)    | 96.8                                 | 0.5<br>(0.2–1.6)                | 11.1               | 0.0                             | 1.92<br>(1.80–<br>2.04)      | 43.6                             |
| Haiku,<br>messy<br>notes,<br>noisy<br>clinician | S2 LLM<br>agent                 | 559 | 85.0<br>(81.8–<br>87.7)    | 97.3                                 | 1.1<br>(0.5–2.3)                | 12.0               | 3.4                             | 0.97<br>(0.87–<br>1.08)      | 10.1                             |
| Haiku,<br>messy<br>notes,<br>noisy<br>clinician | S3<br>Bounds                    | 559 | 85.5<br>(82.3–<br>88.2)    | 96.8                                 | 0.5<br>(0.2–1.6)                | 11.1               | 0.0                             | 1.08<br>(0.97–<br>1.20)      | 0.0                              |

| Condition                                       | System                          | n   | Accuracy,<br>% (95%<br>CI) | Correct<br>when an-<br>swering,<br>% | Under-<br>triage, %<br>(95% CI) | Over-<br>triage, % | Premature<br>commit-<br>ment, % | Questions/cases-<br>(95% CI) | Irrelevant<br>cases-<br>tions, % |
|-------------------------------------------------|---------------------------------|-----|----------------------------|--------------------------------------|---------------------------------|--------------------|---------------------------------|------------------------------|----------------------------------|
| Haiku,<br>messy<br>notes,<br>noisy<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 85.9<br>(82.7–<br>88.5)    | 97.4                                 | 0.5<br>(0.2–1.6)                | 10.7               | 0.0                             | 1.11<br>(1.00–<br>1.23)      | 0.0                              |
| Haiku,<br>messy<br>notes,<br>noisy<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 86.8<br>(83.7–<br>89.3)    | 90.3                                 | 8.6 (6.5–<br>11.2)              | 3.8                | 38.3                            | 0.31<br>(0.25–<br>0.38)      | 0.0                              |
| Qwen,<br>clean<br>notes,<br>ideal<br>clinician  | S1<br>Ask-all                   | 559 | 99.8<br>(99.0–<br>100.0)   | 100.0                                | 0.0<br>(0.0–0.7)                | 0.2                | 0.0                             | 2.15<br>(2.04–<br>2.27)      | 45.5                             |
| Qwen,<br>clean<br>notes,<br>ideal<br>clinician  | S2 LLM<br>agent                 | 559 | 99.6<br>(98.7–<br>99.9)    | 99.8                                 | 0.2<br>(0.0–1.0)                | 0.2                | 0.2                             | 0.92<br>(0.82–<br>1.01)      | 10.1                             |
| Qwen,<br>clean<br>notes,<br>ideal<br>clinician  | S3<br>Bounds                    | 559 | 99.8<br>(99.0–<br>100.0)   | 100.0                                | 0.0<br>(0.0–0.7)                | 0.2                | 0.0                             | 1.17<br>(1.06–<br>1.28)      | 0.0                              |
| Qwen,<br>clean<br>notes,<br>ideal<br>clinician  | S4<br>Bounds+VOI+echo           | 559 | 99.8<br>(99.0–<br>100.0)   | 100.0                                | 0.0<br>(0.0–0.7)                | 0.2                | 0.0                             | 1.14<br>(1.04–<br>1.25)      | 0.0                              |
| Qwen,<br>clean<br>notes,<br>ideal<br>clinician  | S3-bin<br>(miss-<br>ing=normal) | 559 | 91.6<br>(89.0–<br>93.6)    | 91.6                                 | 8.4 (6.4–<br>11.0)              | 0.0                | 46.5                            | 0.25<br>(0.21–<br>0.30)      | 0.0                              |
| Qwen,<br>clean<br>notes,<br>noisy<br>clinician  | S1<br>Ask-all                   | 559 | 84.4<br>(81.2–<br>87.2)    | 97.7                                 | 0.5<br>(0.2–1.6)                | 12.7               | 0.0                             | 2.15<br>(2.04–<br>2.27)      | 44.0                             |
| Qwen,<br>clean<br>notes,<br>noisy<br>clinician  | S2 LLM<br>agent                 | 559 | 84.4<br>(81.2–<br>87.2)    | 97.7                                 | 0.5<br>(0.2–1.6)                | 13.1               | 1.8                             | 0.95<br>(0.84–<br>1.05)      | 8.5                              |

| Condition                                      | System                          | n   | Accuracy,<br>% (95%<br>CI) | Correct<br>when an-<br>swering,<br>% | Under-<br>triage, %<br>(95% CI) | Over-<br>triage, % | Premature<br>commit-<br>ment, % | Questions/cases-<br>(95% CI) | Irrelevant<br>questions, % |
|------------------------------------------------|---------------------------------|-----|----------------------------|--------------------------------------|---------------------------------|--------------------|---------------------------------|------------------------------|----------------------------|
| Qwen,<br>clean<br>notes,<br>noisy<br>clinician | S3<br>Bounds                    | 559 | 84.4<br>(81.2–<br>87.2)    | 97.7                                 | 0.5<br>(0.2–1.6)                | 12.7               | 0.0                             | 1.21<br>(1.09–<br>1.32)      | 0.0                        |
| Qwen,<br>clean<br>notes,<br>noisy<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 84.4<br>(81.2–<br>87.2)    | 97.7                                 | 0.5<br>(0.2–1.6)                | 12.7               | 0.0                             | 1.20<br>(1.09–<br>1.31)      | 0.0                        |
| Qwen,<br>clean<br>notes,<br>noisy<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 87.8<br>(84.9–<br>90.3)    | 91.1                                 | 8.6 (6.5–<br>11.2)              | 3.0                | 45.3                            | 0.26<br>(0.22–<br>0.31)      | 0.0                        |
| Qwen,<br>messy<br>notes,<br>ideal<br>clinician | S1<br>Ask-all                   | 559 | 99.5<br>(98.4–<br>99.8)    | 99.6                                 | 0.2<br>(0.0–1.0)                | 0.4                | 0.0                             | 2.15<br>(2.03–<br>2.28)      | 45.7                       |
| Qwen,<br>messy<br>notes,<br>ideal<br>clinician | S2 LLM<br>agent                 | 559 | 99.8<br>(99.0–<br>100.0)   | 100.0                                | 0.0<br>(0.0–0.7)                | 0.2                | 0.7                             | 0.92<br>(0.82–<br>1.01)      | 10.7                       |
| Qwen,<br>messy<br>notes,<br>ideal<br>clinician | S3<br>Bounds                    | 559 | 99.5<br>(98.4–<br>99.8)    | 99.6                                 | 0.2<br>(0.0–1.0)                | 0.4                | 0.0                             | 1.17<br>(1.06–<br>1.28)      | 0.0                        |
| Qwen,<br>messy<br>notes,<br>ideal<br>clinician | S4<br>Bounds+VOI+echo           | 559 | 99.6<br>(98.7–<br>99.9)    | 99.8                                 | 0.0<br>(0.0–0.7)                | 0.4                | 0.0                             | 1.22<br>(1.10–<br>1.33)      | 0.0                        |
| Qwen,<br>messy<br>notes,<br>ideal<br>clinician | S3-bin<br>(miss-<br>ing=normal) | 559 | 90.0<br>(87.2–<br>92.2)    | 90.0                                 | 9.8 (7.6–<br>12.6)              | 0.2                | 47.6                            | 0.23<br>(0.18–<br>0.27)      | 0.0                        |
| Qwen,<br>messy<br>notes,<br>noisy<br>clinician | S1<br>Ask-all                   | 559 | 84.3<br>(81.0–<br>87.0)    | 97.5                                 | 0.4<br>(0.1–1.3)                | 12.7               | 0.0                             | 2.15<br>(2.03–<br>2.28)      | 43.7                       |



| Condition                          | System                  | n   | Accuracy, % (95% CI) | Correct when answering, % | Under-triage, % (95% CI) | Over-triage, % | Premature commitment, % | Questions/cases- (95% CI) | Irrelevant questions, % |
|------------------------------------|-------------------------|-----|----------------------|---------------------------|--------------------------|----------------|-------------------------|---------------------------|-------------------------|
| Qwen, messy notes, noisy clinician | S2 LLM agent            | 559 | 85.0 (81.8–87.7)     | 97.3                      | 1.1 (0.5–2.3)            | 12.0           | 3.4                     | 0.97 (0.87–1.08)          | 10.1                    |
| Qwen, messy notes, noisy clinician | S3 Bounds               | 559 | 84.3 (81.0–87.0)     | 97.5                      | 0.4 (0.1–1.3)            | 12.7           | 0.0                     | 1.21 (1.09–1.33)          | 0.0                     |
| Qwen, messy notes, noisy clinician | S4 Bounds+VOI+echo      | 559 | 84.3 (81.0–87.0)     | 97.1                      | 0.5 (0.2–1.6)            | 12.7           | 0.0                     | 1.29 (1.17–1.41)          | 0.0                     |
| Qwen, messy notes, noisy clinician | S3-bin (missing=normal) | 559 | 86.6 (83.5–89.2)     | 89.3                      | 10.2 (8.0–13.0)          | 2.7            | 46.7                    | 0.24 (0.19–0.28)          | 0.0                     |

### S7. Paired comparisons (exact McNemar for accuracy, Wilcoxon signed-rank for questions; Holm-adjusted within condition)

Full cohort (n = 1,200):

| Condition            | Comparison                           | Accuracy A vs B, % | Discordant (A only / B only) | McNemar p (Holm) | Questions A vs B | Wilcoxon p (Holm) |
|----------------------|--------------------------------------|--------------------|------------------------------|------------------|------------------|-------------------|
| Oracle, clean, ideal | S3 Bounds vs S1 Ask-all              | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 0.92 vs 1.74     | <0.001            |
| Oracle, clean, ideal | S3 Bounds vs S2 LLM agent            | 99.8 vs 99.6       | 3 / 0                        | 0.750            | 0.92 vs 0.99     | <0.001            |
| Oracle, clean, ideal | S3 Bounds vs S3-bin (missing=normal) | 99.8 vs 91.8       | 99 / 2                       | <0.001           | 0.92 vs 0.21     | <0.001            |
| Oracle, clean, ideal | S4 Bounds+VOI+echo vs S3 Bounds      | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 0.87 vs 0.92     | <0.001            |
| Haiku, clean, ideal  | S3 Bounds vs S1 Ask-all              | 99.4 vs 99.4       | 0 / 0                        | 1.000            | 0.92 vs 1.78     | <0.001            |
| Haiku, clean, ideal  | S3 Bounds vs S2 LLM agent            | 99.4 vs 99.6       | 3 / 5                        | 1.000            | 0.92 vs 0.99     | <0.001            |
| Haiku, clean, ideal  | S3 Bounds vs S3-bin (missing=normal) | 99.4 vs 91.2       | 101 / 2                      | <0.001           | 0.92 vs 0.21     | <0.001            |

| Condition           | Comparison                            | Accuracy A vs B, % | Discordant (A only / B only) | McNemar p (Holm) | Questions A vs B | Wilcoxon p (Holm) |
|---------------------|---------------------------------------|--------------------|------------------------------|------------------|------------------|-------------------|
| Haiku, clean, ideal | S4<br>Bounds+VOI+echo<br>vs S3 Bounds | 99.7 vs 99.4       | 3 / 0                        | 0.750            | 0.90 vs 0.92     | 0.005             |
| Qwen, clean, ideal  | S3 Bounds vs S1 Ask-all               | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 1.23 vs 2.21     | <0.001            |
| Qwen, clean, ideal  | S3 Bounds vs S2 LLM agent             | 99.8 vs 99.6       | 3 / 0                        | 0.750            | 1.23 vs 0.99     | <0.001            |
| Qwen, clean, ideal  | S3 Bounds vs S3-bin (missing=normal)  | 99.8 vs 91.2       | 105 / 2                      | <0.001           | 1.23 vs 0.26     | <0.001            |
| Qwen, clean, ideal  | S4<br>Bounds+VOI+echo<br>vs S3 Bounds | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 1.20 vs 1.23     | 0.004             |
| Haiku, clean, noisy | S3 Bounds vs S1 Ask-all               | 87.0 vs 87.0       | 0 / 0                        | 1.000            | 0.96 vs 1.78     | <0.001            |
| Haiku, clean, noisy | S3 Bounds vs S2 LLM agent             | 87.0 vs 83.5       | 65 / 23                      | <0.001           | 0.96 vs 1.03     | <0.001            |
| Haiku, clean, noisy | S3 Bounds vs S3-bin (missing=normal)  | 87.0 vs 88.0       | 70 / 82                      | 0.750            | 0.96 vs 0.21     | <0.001            |
| Haiku, clean, noisy | S4<br>Bounds+VOI+echo<br>vs S3 Bounds | 87.2 vs 87.0       | 3 / 0                        | 0.750            | 0.94 vs 0.96     | 0.078             |

Paired subset (n = 559):

| Condition                           | Comparison                            | Accuracy A vs B, % | Discordant (A only / B only) | McNemar p (Holm) | Questions A vs B | Wilcoxon p (Holm) |
|-------------------------------------|---------------------------------------|--------------------|------------------------------|------------------|------------------|-------------------|
| Haiku, clean notes, ideal clinician | S3 Bounds vs S1 Ask-all               | 99.1 vs 99.1       | 0 / 0                        | 1.000            | 0.85 vs 1.71     | <0.001            |
| Haiku, clean notes, ideal clinician | S3 Bounds vs S2 LLM agent             | 99.1 vs 99.6       | 1 / 4                        | 0.750            | 0.85 vs 0.92     | 0.002             |
| Haiku, clean notes, ideal clinician | S3 Bounds vs S3-bin (missing=normal)  | 99.1 vs 91.4       | 45 / 2                       | <0.001           | 0.85 vs 0.20     | <0.001            |
| Haiku, clean notes, ideal clinician | S4<br>Bounds+VOI+echo<br>vs S3 Bounds | 99.6 vs 99.1       | 3 / 0                        | 0.750            | 0.82 vs 0.85     | 0.023             |
| Haiku, clean notes, noisy clinician | S3 Bounds vs S1 Ask-all               | 88.2 vs 88.2       | 0 / 0                        | 1.000            | 0.88 vs 1.71     | <0.001            |
| Haiku, clean notes, noisy clinician | S3 Bounds vs S2 LLM agent             | 88.2 vs 84.4       | 34 / 13                      | 0.012            | 0.88 vs 0.95     | <0.001            |

| Condition                           | Comparison                           | Accuracy A vs B, % | Discordant (A only / B only) | McNemar p (Holm) | Questions A vs B | Wilcoxon p (Holm) |
|-------------------------------------|--------------------------------------|--------------------|------------------------------|------------------|------------------|-------------------|
| Haiku, clean notes, noisy clinician | S3 Bounds vs S3-bin (missing=normal) | 88.2 vs 88.7       | 31 / 34                      | 1.000            | 0.88 vs 0.20     | <0.001            |
| Haiku, clean notes, noisy clinician | S4 Bounds+VOI+echo vs S3 Bounds      | 88.7 vs 88.2       | 3 / 0                        | 0.750            | 0.86 vs 0.88     | 0.342             |
| Haiku, messy notes, ideal clinician | S3 Bounds vs S1 Ask-all              | 98.6 vs 98.6       | 0 / 0                        | 1.000            | 1.04 vs 1.92     | <0.001            |
| Haiku, messy notes, ideal clinician | S3 Bounds vs S2 LLM agent            | 98.6 vs 99.8       | 0 / 7                        | 0.047            | 1.04 vs 0.92     | 0.002             |
| Haiku, messy notes, ideal clinician | S3 Bounds vs S3-bin (missing=normal) | 98.6 vs 90.7       | 46 / 2                       | <0.001           | 1.04 vs 0.30     | <0.001            |
| Haiku, messy notes, ideal clinician | S4 Bounds+VOI+echo vs S3 Bounds      | 99.1 vs 98.6       | 3 / 0                        | 0.500            | 1.04 vs 1.04     | 0.933             |
| Haiku, messy notes, noisy clinician | S3 Bounds vs S1 Ask-all              | 85.5 vs 85.5       | 0 / 0                        | 1.000            | 1.08 vs 1.92     | <0.001            |
| Haiku, messy notes, noisy clinician | S3 Bounds vs S2 LLM agent            | 85.5 vs 85.0       | 30 / 27                      | 1.000            | 1.08 vs 0.97     | 0.032             |
| Haiku, messy notes, noisy clinician | S3 Bounds vs S3-bin (missing=normal) | 85.5 vs 86.8       | 33 / 40                      | 1.000            | 1.08 vs 0.31     | <0.001            |
| Haiku, messy notes, noisy clinician | S4 Bounds+VOI+echo vs S3 Bounds      | 85.9 vs 85.5       | 3 / 1                        | 1.000            | 1.11 vs 1.08     | 0.048             |
| Qwen, clean notes, ideal clinician  | S3 Bounds vs S1 Ask-all              | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 1.17 vs 2.15     | <0.001            |
| Qwen, clean notes, ideal clinician  | S3 Bounds vs S2 LLM agent            | 99.8 vs 99.6       | 1 / 0                        | 1.000            | 1.17 vs 0.92     | <0.001            |
| Qwen, clean notes, ideal clinician  | S3 Bounds vs S3-bin (missing=normal) | 99.8 vs 91.6       | 47 / 1                       | <0.001           | 1.17 vs 0.25     | <0.001            |
| Qwen, clean notes, ideal clinician  | S4 Bounds+VOI+echo vs S3 Bounds      | 99.8 vs 99.8       | 0 / 0                        | 1.000            | 1.14 vs 1.17     | 0.033             |
| Qwen, clean notes, noisy clinician  | S3 Bounds vs S1 Ask-all              | 84.4 vs 84.4       | 0 / 0                        | 1.000            | 1.21 vs 2.15     | <0.001            |
| Qwen, clean notes, noisy clinician  | S3 Bounds vs S2 LLM agent            | 84.4 vs 84.4       | 32 / 32                      | 1.000            | 1.21 vs 0.95     | <0.001            |

| Condition                          | Comparison                           | Accuracy A vs B, % | Discordant (A only / B only) | McNemar p (Holm) | Questions A vs B | Wilcoxon p (Holm) |
|------------------------------------|--------------------------------------|--------------------|------------------------------|------------------|------------------|-------------------|
| Qwen, clean notes, noisy clinician | S3 Bounds vs S3-bin (missing=normal) | 84.4 vs 87.8       | 34 / 53                      | 0.212            | 1.21 vs 0.26     | <0.001            |
| Qwen, clean notes, noisy clinician | S4                                   | 84.4 vs 84.4       | 0 / 0                        | 1.000            | 1.20 vs 1.21     | 0.960             |
| Qwen, messy notes, ideal clinician | Bounds+VOI+echo vs S3 Bounds         | 99.5 vs 99.5       | 0 / 0                        | 1.000            | 1.17 vs 2.15     | <0.001            |
| Qwen, messy notes, ideal clinician | S3 Bounds vs S1 Ask-all              | 99.5 vs 99.8       | 0 / 2                        | 1.000            | 1.17 vs 0.92     | <0.001            |
| Qwen, messy notes, ideal clinician | S3 Bounds vs S3-bin (missing=normal) | 99.5 vs 90.0       | 54 / 1                       | <0.001           | 1.17 vs 0.23     | <0.001            |
| Qwen, messy notes, ideal clinician | S4                                   | 99.6 vs 99.5       | 1 / 0                        | 1.000            | 1.22 vs 1.17     | 0.006             |
| Qwen, messy notes, ideal clinician | Bounds+VOI+echo vs S3 Bounds         | 99.6 vs 99.5       | 1 / 0                        | 1.000            | 1.22 vs 1.17     | 0.006             |
| Qwen, messy notes, noisy clinician | S3 Bounds vs S1 Ask-all              | 84.3 vs 84.3       | 0 / 0                        | 1.000            | 1.21 vs 2.15     | <0.001            |
| Qwen, messy notes, noisy clinician | S3 Bounds vs S2 LLM agent            | 84.3 vs 85.0       | 31 / 35                      | 1.000            | 1.21 vs 0.97     | <0.001            |
| Qwen, messy notes, noisy clinician | S3 Bounds vs S3-bin (missing=normal) | 84.3 vs 86.6       | 42 / 55                      | 0.892            | 1.21 vs 0.24     | <0.001            |
| Qwen, messy notes, noisy clinician | S4                                   | 84.3 vs 84.3       | 2 / 2                        | 1.000            | 1.29 vs 1.21     | <0.001            |
| Qwen, messy notes, noisy clinician | Bounds+VOI+echo vs S3 Bounds         | 84.3 vs 84.3       | 2 / 2                        | 1.000            | 1.29 vs 1.21     | <0.001            |

## S8. Real case reports, all splits and extractors

| Split | Extractor | Calculator      | n   | Code reproduces label, % | Determined from note, % (95% CI) | Truth still possible, % | Category correct, missing=normal, % | Entity agreement, % |
|-------|-----------|-----------------|-----|--------------------------|----------------------------------|-------------------------|-------------------------------------|---------------------|
| train | Haiku 4.5 | Cockcroft-Gault | 125 | 51                       | 88.0 (81.1–92.6)                 | 85                      | 71                                  | 90                  |
| train | Haiku 4.5 | CURB-65         | 85  | 100                      | 49.4 (39.0–59.8)                 | 93                      | 82                                  | 89                  |
| train | Haiku 4.5 | HEART           | 125 | 100                      | 12.8 (8.0–19.8)                  | 90                      | 22                                  | 73                  |

| Split | Extractor      | Calculator          | n   | Code re-<br>produces<br>label, % | Determined<br>from note,<br>% (95%<br>CI) | Truth still<br>possible,<br>% | Category<br>correct,<br>miss-<br>ing=normal,<br>% | Entity<br>agree-<br>ment, % |
|-------|----------------|---------------------|-----|----------------------------------|-------------------------------------------|-------------------------------|---------------------------------------------------|-----------------------------|
| train | Haiku 4.5      | PERC                | 124 | 100                              | 90.3<br>(83.8–<br>94.4)                   | 100                           | 94                                                | 90                          |
| train | Haiku 4.5      | Wells PE            | 125 | 100                              | 19.2<br>(13.3–<br>27.0)                   | 98                            | 81                                                | 87                          |
| train | Haiku 4.5      | All                 | 584 | 90                               | 52.1<br>(48.0–<br>56.1)                   | 93                            | 69                                                | 86                          |
| train | Qwen3.5-<br>9B | Cockcroft-<br>Gault | 125 | 51                               | 84.8<br>(77.5–<br>90.0)                   | 90                            | 72                                                | 92                          |
| train | Qwen3.5-<br>9B | CURB-65             | 85  | 100                              | 41.2<br>(31.3–<br>51.8)                   | 95                            | 81                                                | 92                          |
| train | Qwen3.5-<br>9B | HEART               | 125 | 100                              | 12.0<br>(7.4–18.9)                        | 93                            | 9                                                 | 74                          |
| train | Qwen3.5-<br>9B | PERC                | 124 | 100                              | 87.9<br>(81.0–<br>92.5)                   | 100                           | 91                                                | 89                          |
| train | Qwen3.5-<br>9B | Wells PE            | 125 | 100                              | 9.6<br>(5.6–16.0)                         | 99                            | 78                                                | 88                          |
| train | Qwen3.5-<br>9B | All                 | 584 | 90                               | 47.4<br>(43.4–<br>51.5)                   | 95                            | 65                                                | 87                          |
| test  | Haiku 4.5      | Cockcroft-<br>Gault | 20  | 40                               | 95.0<br>(76.4–<br>99.1)                   | 100                           | 95                                                | 96                          |
| test  | Haiku 4.5      | CURB-65             | 20  | 100                              | 55.0<br>(34.2–<br>74.2)                   | 100                           | 100                                               | 97                          |
| test  | Haiku 4.5      | HEART               | 20  | 95                               | 60.0<br>(38.7–<br>78.1)                   | 85                            | 45                                                | 87                          |
| test  | Haiku 4.5      | PERC                | 20  | 100                              | 75.0<br>(53.1–<br>88.8)                   | 100                           | 95                                                | 99                          |
| test  | Haiku 4.5      | Wells PE            | 20  | 100                              | 10.0<br>(2.8–30.1)                        | 100                           | 85                                                | 89                          |
| test  | Haiku 4.5      | All                 | 100 | 87                               | 59.0<br>(49.2–<br>68.1)                   | 97                            | 84                                                | 94                          |
| test  | Qwen3.5-<br>9B | Cockcroft-<br>Gault | 20  | 40                               | 90.0<br>(69.9–<br>97.2)                   | 100                           | 90                                                | 96                          |

| Split | Extractor  | Calculator | n   | Code re-<br>produces<br>label, % | Determined<br>from note,<br>% (95%<br>CI) | Truth still<br>possible,<br>% | Category<br>correct,<br>miss-<br>ing=normal,<br>% | Entity<br>agree-<br>ment, % |
|-------|------------|------------|-----|----------------------------------|-------------------------------------------|-------------------------------|---------------------------------------------------|-----------------------------|
| test  | Qwen3.5-9B | CURB-65    | 20  | 100                              | 50.0<br>(29.9–70.1)                       | 95                            | 90                                                | 98                          |
| test  | Qwen3.5-9B | HEART      | 20  | 95                               | 30.0<br>(14.5–51.9)                       | 95                            | 35                                                | 77                          |
| test  | Qwen3.5-9B | PERC       | 20  | 100                              | 75.0<br>(53.1–88.8)                       | 100                           | 95                                                | 96                          |
| test  | Qwen3.5-9B | Wells PE   | 20  | 100                              | 10.0<br>(2.8–30.1)                        | 100                           | 85                                                | 89                          |
| test  | Qwen3.5-9B | All        | 100 | 87                               | 51.0<br>(41.3–60.6)                       | 98                            | 79                                                | 91                          |

## S10. Sensitivity to implausible synthetic cases

Characteristics were sampled independently, so some cases are clinically implausible. We flagged established atherosclerotic disease with no risk factors, and age under 40 with three or more risk factors or atherosclerotic disease. We then recomputed the main results without those cases (clean notes, ideal clinician).

12 of 1200 cases flagged: heart-0017 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0023 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0036 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0069 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0087 (established atherosclerotic disease with no risk factors); heart-0104 (established atherosclerotic disease with no risk factors); heart-0128 (established atherosclerotic disease with no risk factors); heart-0132 (established atherosclerotic disease with no risk factors); heart-0135 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0137 (established atherosclerotic disease with no risk factors); heart-0172 (age < 40 with  $\geq 3$  risk factors or atherosclerotic disease); heart-0195 (established atherosclerotic disease with no risk factors)

| Extraction | Cases                | System                       | n    | Accuracy, % | Under-triage,<br>% | Questions/case |
|------------|----------------------|------------------------------|------|-------------|--------------------|----------------|
| Oracle     | all                  | S1 Ask-all                   | 1200 | 99.8        | 0.0                | 1.74           |
| Oracle     | all                  | S2 LLM agent                 | 1200 | 99.6        | 0.1                | 0.99           |
| Oracle     | all                  | S3 Bounds                    | 1200 | 99.8        | 0.0                | 0.92           |
| Oracle     | all                  | S4                           | 1200 | 99.8        | 0.0                | 0.87           |
| Oracle     | all                  | Bounds+VOI+echo              |      |             |                    |                |
| Oracle     | all                  | S3-bin (miss-<br>ing=normal) | 1200 | 91.8        | 8.2                | 0.21           |
| Oracle     | excluding<br>flagged | S1 Ask-all                   | 1188 | 99.8        | 0.0                | 1.73           |
| Oracle     | excluding<br>flagged | S2 LLM agent                 | 1188 | 99.6        | 0.1                | 0.99           |
| Oracle     | excluding<br>flagged | S3 Bounds                    | 1188 | 99.8        | 0.0                | 0.92           |

| Extraction | Cases                | System                       | n    | Accuracy, % | Under-triage,<br>% | Questions/case |
|------------|----------------------|------------------------------|------|-------------|--------------------|----------------|
| Oracle     | excluding<br>flagged | S4<br>Bounds+VOI+echo        | 1188 | 99.8        | 0.0                | 0.87           |
| Oracle     | excluding<br>flagged | S3-bin (miss-<br>ing=normal) | 1188 | 91.9        | 8.1                | 0.21           |
| Haiku 4.5  | all                  | S1 Ask-all                   | 1200 | 99.4        | 0.1                | 1.78           |
| Haiku 4.5  | all                  | S2 LLM agent                 | 1200 | 99.6        | 0.1                | 0.99           |
| Haiku 4.5  | all                  | S3 Bounds                    | 1200 | 99.4        | 0.1                | 0.92           |
| Haiku 4.5  | all                  | S4<br>Bounds+VOI+echo        | 1200 | 99.7        | 0.1                | 0.90           |
| Haiku 4.5  | all                  | S3-bin (miss-<br>ing=normal) | 1200 | 91.2        | 8.5                | 0.21           |
| Haiku 4.5  | excluding<br>flagged | S1 Ask-all                   | 1188 | 99.4        | 0.1                | 1.77           |
| Haiku 4.5  | excluding<br>flagged | S2 LLM agent                 | 1188 | 99.6        | 0.1                | 0.99           |
| Haiku 4.5  | excluding<br>flagged | S3 Bounds                    | 1188 | 99.4        | 0.1                | 0.93           |
| Haiku 4.5  | excluding<br>flagged | S4<br>Bounds+VOI+echo        | 1188 | 99.7        | 0.1                | 0.90           |
| Haiku 4.5  | excluding<br>flagged | S3-bin (miss-<br>ing=normal) | 1188 | 91.3        | 8.3                | 0.21           |
| Qwen3.5-9B | all                  | S1 Ask-all                   | 1200 | 99.8        | 0.0                | 2.21           |
| Qwen3.5-9B | all                  | S2 LLM agent                 | 1200 | 99.6        | 0.1                | 0.99           |
| Qwen3.5-9B | all                  | S3 Bounds                    | 1200 | 99.8        | 0.0                | 1.23           |
| Qwen3.5-9B | all                  | S4<br>Bounds+VOI+echo        | 1200 | 99.8        | 0.0                | 1.20           |
| Qwen3.5-9B | all                  | S3-bin (miss-<br>ing=normal) | 1200 | 91.2        | 8.8                | 0.26           |
| Qwen3.5-9B | excluding<br>flagged | S1 Ask-all                   | 1188 | 99.8        | 0.0                | 2.20           |
| Qwen3.5-9B | excluding<br>flagged | S2 LLM agent                 | 1188 | 99.6        | 0.1                | 0.99           |
| Qwen3.5-9B | excluding<br>flagged | S3 Bounds                    | 1188 | 99.8        | 0.0                | 1.23           |
| Qwen3.5-9B | excluding<br>flagged | S4<br>Bounds+VOI+echo        | 1188 | 99.8        | 0.0                | 1.20           |
| Qwen3.5-9B | excluding<br>flagged | S3-bin (miss-<br>ing=normal) | 1188 | 91.4        | 8.6                | 0.26           |

## S9. Exact prompts

Prompt versions: renderer v3, extractor x1, agent a1. The example fact sheet is for one CURB-65 case with a trap; <the rendered note> stands for the note text.

### Renderer (system)

You write realistic clinical notes for a research dataset. Every patient is fictional; the notes a

## Renderer (user), example CURB-65 case

Write one clinical note in US English (en-US).

Setting: Emergency department. Adult with suspected community-acquired pneumonia.

Style guide:

- US emergency department or inpatient documentation conventions.
- Typical sections: Chief complaint, HPI, PMH, Medications, Allergies, Social history, Exam (vitals first), Results, Assessment and plan. Not every section is required.
- Common US abbreviations are fine (HPI, PMH, SOB, CP, ROS, HR, RR, BP, SpO2, NKDA).
- Units: US conventional (mg/dL, lb or kg, °F or °C) except where the fact sheet gives a unit.
- Terse clinical style; sentence fragments are normal.

Fact sheet (each item is mandatory):

- [confusion] STATE: the patient is alert and fully oriented, not confused.
- [urea] STATE (number exactly): serum urea 8 mmol/L. Also mention a Serum urea (mmol/L) of 6.6 fr
- [resp\_rate] DO NOT MENTION Respiratory rate (/min). Include nothing from which it could be infer
- [sbp] DO NOT MENTION Systolic blood pressure (mmHg). Include nothing from which it could be infe
- [dbp] DO NOT MENTION Diastolic blood pressure (mmHg). Include nothing from which it could be inf
- [age] DO NOT MENTION Age (years). Include nothing from which it could be inferred (no value, no

Rules:

- Write numbers exactly as given in the fact sheet (same digits and decimals), with the unit given
- Express each item in natural clinical language, as a clinician would chart it; do not copy the f
- Every STATE item must appear explicitly in the note, including negative ones (e.g. "no diabetes"
- Follow each item exactly. Do not add information about any fact-sheet item beyond what it says,
- No blanket or complete statements that would cover a DO NOT MENTION item: no "Medications: none"
- You may add other realistic details (presenting complaint, unrelated history, unrelated examinatio
- Do not name any clinical score or rule (HEART score, CURB-65, CURB65, qSOFA, PERC, Wells, Cockcr
- Do not refer to these instructions, the fact sheet, or the note being synthetic.
- Output only the note, 150–350 words, plain text (simple section headings allowed).

## Judge (system)

You audit synthetic clinical notes against a list of parameters. For each parameter, decide only f

## Judge (user)

Note:

```
<note>  
<the rendered note>  
</note>
```

Parameters:

- confusion: Confusion (AMT ≤ 8 or new disorientation to person/place/time) [yes/no finding]
- urea: Serum urea (mmol/L) [numeric (mmol/L)]
- resp\_rate: Respiratory rate (/min) [numeric (/min)]
- sbp: Systolic blood pressure (mmHg) [numeric (mmHg)]
- dbp: Diastolic blood pressure (mmHg) [numeric (mmHg)]
- age: Age (years) [numeric (years)]

For every parameter, return one object:



`{"param": "<id>", "status": "stated" | "negated_or_normal" | "not_mentioned" | "implied", "level":`

- stated: the value or finding is explicitly documented (for yes/no findings: present).
- negated\_or\_normal: explicitly absent, denied, or documented as normal without a value.
- not\_mentioned: nothing in the note documents or implies it.
- implied: not explicitly documented, but inferable from the note.

Return a JSON array with exactly one object per parameter, in the order listed, and nothing else.

### **Extractor (system)**

You extract clinical parameters from a clinical note into a fixed JSON structure. You report only

### **Extractor (user)**

Note:

`<note>`

`<the rendered note>`

`</note>`

Parameters:

- confusion: Confusion (AMT  $\leq$  8 or new disorientation to person/place/time) [yes/no finding]
- urea: Serum urea (mmol/L) [numeric; unit tokens: mmol/L, urea\_mg/dL, bun\_mg/dL]
- resp\_rate: Respiratory rate (/min) [numeric; unit tokens: /min]
- sbp: Systolic blood pressure (mmHg) [numeric; unit tokens: mmHg]
- dbp: Diastolic blood pressure (mmHg) [numeric; unit tokens: mmHg]
- age: Age (years) [numeric; unit tokens: years]

For every parameter give:

- "status":
  - "present": the note states the finding, or states a value/level for it.
  - "absent": the note explicitly says the finding is absent or denied, or that the parameter is n
  - "unknown": the note does not document it. Do not infer absence from silence: if the note does
- "value": for yes/no findings use null; for graded parameters one of the listed levels; for numer
- "unit": for numeric parameters, the unit as written, mapped to one of the listed unit tokens; ot
- "evidence": the shortest exact quote from the note (copied character for character) that support
- "confidence": your probability (0 to 1) that status and value are correct.

If the note gives several values for the same parameter (e.g. an earlier reading or a previous vis

### **S2 agent (system)**

You are a clinical decision-support agent working with an emergency clinician. You determine a pat

### **S2 agent (first user turn)**

Calculator: CURB-65

Decision categories: low, moderate, high

Clinical note:

`<note>`

`<the rendered note>`

`</note>`

Calculator parameters (ids you can ask about or pass to calculate):

- confusion: Confusion (AMT  $\leq$  8 or new disorientation to person/place/time) [yes/no]
- urea: Serum urea (mmol/L) [number in mmol/L]
- resp\_rate: Respiratory rate (/min) [number in /min]
- sbp: Systolic blood pressure (mmHg) [number in mmHg]
- dbp: Diastolic blood pressure (mmHg) [number in mmHg]
- age: Age (years) [number in years]

Actions (reply with one JSON object per turn):

1. {"action": "ask", "parameter": "<id>", "question": "<your question to the clinician>"}  
The clinician answers from the patient (or says the value is not available).
2. {"action": "calculate", "values": {"<id>": <value>, ...}}  
Runs the calculator. Give every parameter: yes/no as true/false, graded ones as the level name,
3. {"action": "answer", "category": "<one of: low, moderate, high, cannot\_determine>"}  
Your final decision category. Use "cannot\_determine" only if it truly cannot be determined.